

Week 13: Where the Gap Comes From

Hankel_AD and Hankel_CDC_AD, 23–30 September

Benji Clothier

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The gap to SOTA was **one dataset family**.

Locality fixes it.

Local tangent geometry (CDC) did not beat plain distance.

Corrections to earlier claims

- “max_d cap ties unclamped” (Week 12) – **false**
- Golyandina & Zhigljavsky (2011) – **does not exist**
- “1.8 points behind SOTA” – **misleading framing**

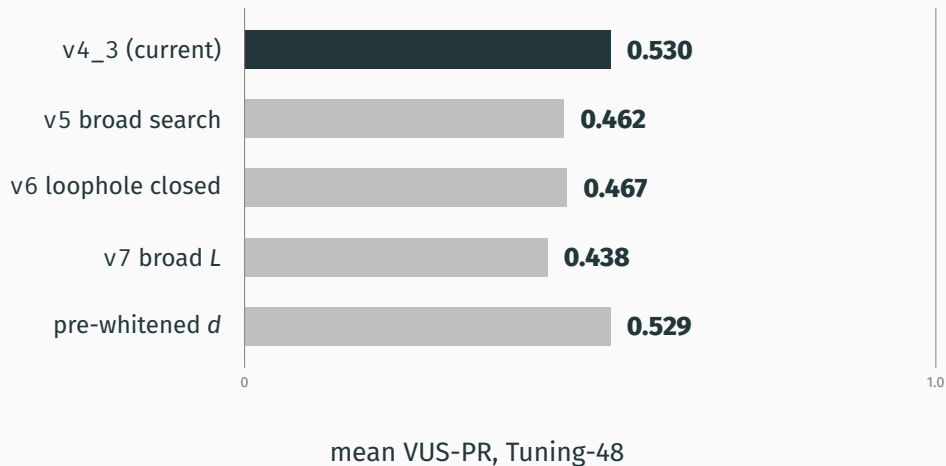
Hankel_AD

How much is left on the table?



mean VUS-PR, Eva · oracle picks $d \approx 50$, we pick $d \approx 11$

Four redesigns of (L, d) , four failures



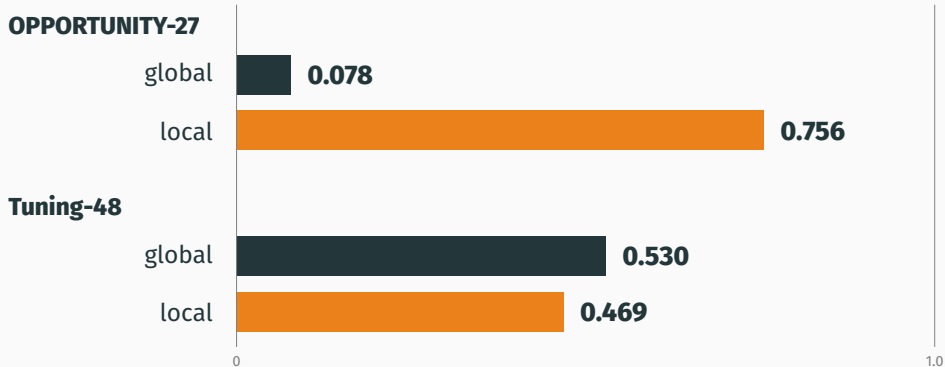
The gap is one family

OPPORTUNITY (HumanActivity) – 27 of 350 datasets



27 / 27 lose · **excluding them:** $-0.020 \rightarrow +0.034$

Locality fixes it – a straight swap breaks the rest



Ensemble: take the more confident scorer, per window

$$z_s(w) = \frac{\text{score}_s(w) - \text{median}(\text{score}_s[\text{train}])}{1.4826 \cdot \text{MAD}(\text{score}_s[\text{train} \cup \text{test}])}, \quad s \in \{\text{global}, \text{local}\}$$

$$\text{ensemble}(w) = \max(z_{\text{global}}(w), z_{\text{local}}(w))$$

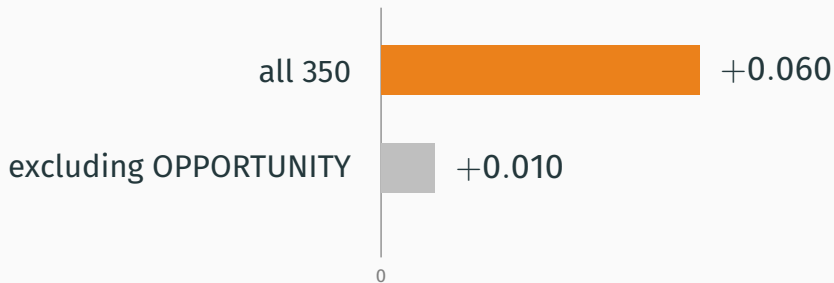
no new tuned constants · label-free

Official Eva split ($n = 350$)



mean VUS-PR · vs. global: 138 wins, 74 losses · Wilcoxon $p = 5.4 \times 10^{-5}$

Where the gain comes from



ensemble – global, mean VUS-PR · excluded subset: Wilcoxon $p = 0.051$

- Not blind: OPPORTUNITY is inside the Eva split
- Most of the gain is that one family
- SOTA figure is a reference, not a rerun
- z-score scale uses unlabelled test-period scores
- 7 Tuning datasets still lose badly to global
- 4 OPPORTUNITY datasets fail under every local method

Hankel_CDC_AD

Synthetic ground truth: tangents exact, curvature wrong

	tangent-plane error	curvature (true = 1)
sphere, $n = 300$	≤ 0.0015	4.2
sphere, $n = 2000$	≤ 0.0002	24.0
sphere, $n = 5000$	≤ 0.0002	59.1

tangent score: AUC 1.0 on synthetic off-manifold anomalies

Real data: CDC does not beat plain distance



Tuning: 14 wins – 14 losses – 2 ties ($p = 0.927$)

Fixed

- GPU port: $8.5\times$ faster
- three memory bottlenecks
- scoring new points

Open

- results differ between sessions
- caps move VUS-PR by up to 0.33
- n_{train} capped at 4000

- Blind re-validation of the ensemble
- Decide on a guarded rule for the 7 failures
- Cap-free CDC reruns before quoting its aggregates
- Finish the citation clean-up